AI Usage Disclosure

AI Usage Philosophy

AI tools were used as development assistants rather than decision makers throughout this project. While AI assisted with code generation, debugging, documentation, and analytical approaches, all technical decisions, validation procedures, business interpretations, and final conclusions were made by the author.

All AI-generated outputs were critically reviewed, tested, and modified where necessary before inclusion in the final submission. AI suggestions were treated as recommendations rather than authoritative answers, and their suitability was evaluated against project requirements, data characteristics, and business objectives.

The final deliverables represent human-led work supported by AI-assisted productivity and problem-solving tools.

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1. AI Tools Used

Tool	Version	Purpose	Usage Frequency
DeepSeek	Latest	Code structure, debugging, report template	High
Claude	Claude 3.5 Sonnet	Pipeline design, statistical explanation	High
GitHub Copilot	Latest	Code completion, function generation	Medium

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2. AI-Assisted Sections

Data Engineering

- Assisted with ingestion pipeline structure
- Suggested error handling approaches
- Helped debug SQLite loading issues
- Assisted in schema design

Data Analysis

- Suggested EDA approaches
- Assisted with statistical test implementation
- Helped interpret cross-city comparisons

Reporting

- Assisted with report organization
- Suggested wording for business insights
- Helped improve documentation clarity

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3. Key Prompts Used

Data Engineering Prompts

1. *"Design error handling and logging for reusable Airbnb data ingestion pipeline"*

   Response: Provided structure with LocalDataLoader class

   Modification: Added Windows compatibility (encoding fixes)

2. *"Implement star schema for Airbnb data with SQLite"*

   Response: Generated dimension and fact table creation

   Modification: Added dynamic column detection

Statistical Prompts

1. *"Implement hypothesis tests H1-H5 with effect sizes and business interpretations"*

   Response: Provided scipy implementations

Modification: Added Levene's test for variance checking

2. *"Create correlation analysis with VIF for multicollinearity"*

   Response: Generated correlation matrix code

   Modification: Added business interpretation

ML Prompts

1. *"Implement XGBoost price prediction with feature importance"*

   Response: Provided model training code

   Modification: Added SHAP for explainability

NLP Prompts

1. *"Create sentiment analysis and topic modeling for Airbnb reviews"*

Response: Provided VADER and NMF implementation

Modification: Added RAG system

Dashboard Prompts

1. *"Create Streamlit dashboard for cross-city Airbnb comparison"*

   Response: Provided dashboard structure

   Modification: Added radar chart for comparison

Report Prompts

1. *"Write business interpretations for EDA findings"*

   Response: Provided initial drafts

   Modification: Customized for three-city analysis

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4. Output Validation

Code Validation

- **All AI-generated code**: Reviewed line-by-line
- **Error handling**: Verified for edge cases
- **Data types**: Validated for SQLite compatibility
- **Performance**: Tested with sample datasets

Statistical Validation

- **Manual Cross-Check**: Key calculations verified with alternative methods
- **Assumption Checks**: Normality, variance, independence verified
- **Effect Sizes**: Compared with literature benchmarks

Model Validation

- **Cross-Validation**: 5-fold CV to prevent overfitting
- **Residual Analysis**: Checked for patterns

- **Feature Importance**: Validated against domain knowledge

Business Validation

- **Interpretations**: Verified against known market patterns
- **Recommendations**: Assessed for feasibility
- **Insights**: Cross-referenced with multiple sources

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5. Modifications Made

Code Modifications

AI Output	Modification	Reason
DuckDB implementation	Converted to SQLite	Windows compatibility
Standard emojis	Removed from logging	Console encoding issues

AI Output	Modification	Reason
Hardcoded paths	Converted to config	Reusability
Price tier logic	Customized function	Null handling

Analysis Modifications

AI Output	Modification	Reason
Generic interpretations	Customized for 3 cities	Business context
Standard visualizations	Enhanced for accessibility	Clarity
One-size-fits-all	City-specific breakdowns	Comparative analysis

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6. Critical Assessment

Rejected AI Suggestions

1. **Neural Network Architecture Suggestion**: Use deep learning for price prediction

   Rejection Reason: Overkill for this dataset, XGBoost performed well

   Decision: Stick with XGBoost

2. **Cloud Deployment Suggestion**: Deploy to AWS/GCP

   Rejection Reason: Assessment focus on analysis, not deployment

   Decision: Document architecture instead

3. **Complex Feature Engineering Suggestion**: Create 30+ interaction features

   Rejection Reason: Risk of overfitting

   Decision: Keep to 15 well-chosen features

Substantially Modified AI Suggestions

1. **Price Tier Assignment Original**: Used `pd.cut()` with bins

   Modified: Custom function with null handling

   Reason: Avoided `NoneType` comparison errors

2. **Sentiment Analysis Original**: Only used VADER

   Modified: Added correlation with ratings

   Reason: Business value of sentiment-rating relationship

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7. Lessons Learned

Effective AI Collaboration

1. **Specific Prompts**: More specific prompts yield better outputs
2. **Iterative Refinement**: Multiple iterations improve quality

3. **Validation is Essential**: Always verify AI outputs
4. **Context Matters**: Provide business context in prompts
5. **Document Everything**: Track what was AI-assisted

AI Limitations

1. **Context Window**: Cannot process full dataset at once
2. **Business Nuance**: Requires human interpretation
3. **Edge Cases**: Often missed in initial implementation
4. **Current Events**: May not know latest data
5. **Customization**: Always needs adaptation

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8. Ethical Considerations

Responsible AI Use

1. **Disclosure**: Full transparency about AI usage
2. **Validation**: All outputs verified
3. **Ownership**: Intellectual ownership maintained
4. **Originality**: AI used for acceleration, not replacement
5. **Attribution**: Not claiming AI work as original

Data Privacy

1. **No Personal Data**: All data is public Airbnb data
2. **No API Keys**: Not exposed in code
3. **Local Processing**: All computation local
4. **GDPR Compliance**: Anonymized data only

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9. AI Contribution Summary

Area	AI Contribution	Human Contribution	Balance
Code Development	40%	60%	Human-led
Analysis	30%	70%	Human-led
Interpretation	20%	80%	Human-led
Report Writing	30%	70%	Human-led
Overall	**30%**	**70%**	**Human-led**

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10. Conclusion

AI tools were used responsibly as acceleration and assistance tools throughout this assessment. All AI-generated outputs were critically evaluated, validated, and modified as needed. The final submission represents human-led work with AI assistance, maintaining intellectual ownership and professional integrity.

Key Principles Followed:

- ✅ Full disclosure
- ✅ Critical evaluation
- ✅ Output validation
- ✅ Human leadership
- ✅ Business context
